- Decision Trees
- Sklearn ?? → sprote
  - Lotsa parameters → imbalanced data
  - → sklearn
- → Splitting of numerical features
- → Underfitting & overfitting
- → Practical reasoning / function what ??

Imbalanced data

① subsample (undersample)

→ 1% subsample → time ↑

(9,800) (200)

② class - weights

by loss $x|w$

$w_0 = 1000$, $w_1 = 1$

$\Rightarrow$ Imp(1) = Imp(0)

weight → (1000) → $\frac{1}{1000}$

$\frac{1}{1000} \times 1000 = 1$

③ under - sample

1000 data points

pick 150 data points at random

SMOTE (Synthetic Minority Oversampling Technique)

① pick some point x_i & pick some k points

② pick a point x_i & pick some k points

① random value $C(x_i)$ by 0.2

$\Rightarrow$ $x_{new} = x_i(1-\alpha) + \alpha \times x_i(C(x_i))$

Decision Trees

Numerical feature

for (categorical) ∈ classify

for C... ...

→ ??

② best gain feature (numerical) by g_k

	f_1	f_2	y
1	1.5	2.5	+
2	1.7	3.1	+
3	1.9	4.0	+
4	2.0	5.2	−
5	2.1	6.3	−
6	2.3	7.1	+

C_k

① each and every threshold → calculate impurity

① $C_1 = 1.5$ 0.2 x

② $C_2 = 1.7$ 0.2 x

③ $C_3 = 2.1$ 0.2 x

⋮ → C_6 (Minimum)

Hack → bin your feature

age < C_{age} (20-40) (40-70) → (70-90)

overfitting v/s underfitting

leaf nodes are pure

→ $\frac{C_{leaf}}{C_{total}}$ → 100% accuracy (validation)

underfitting → don't split at all

overfitting → a lot of split

depth is too large

Underfitting (Children tree)

Overfitting (deep tree) (a lot of splits)

depth 2 (Decision stump)

depth → hyper parameter

stopping conditions

① pure node (overfitted)

② max - depth → 5 to 8

③ minimum point impurity to split function

min - impurity < 0.01

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How DT works from Geometric view

(non - linear)

Decision boundary (axis parallel lines) (hyperplane)

Impact of attributes

weighting

knn → depends → k & my point → y

↳ else → y

DT → depth → large → yes

↳ shallow → No

Standardization / feature scaling

Decision Trees → No standardization

→ Threshold (single feature)

↳ Category → ✓ IG

Encoding (categorical features)

cat features → No encoding

→ OHE → Target encoding

education level → not (C)

→ with (C)

→ bachelors (C)

→ not bachelor (C)

→ PhD (C)

very high dimension

→ data point x d (sparse)

$d \uparrow \uparrow \uparrow$

DT → drop → not performed

→ Random forest

→ Gradient boost DT

Imbalance of data (class balance is very)

y = 99%]

y = 1%]

↳ class - weight

↳ over / under sample

Time & space complexity

space ⇒ # number of nodes → m

→ O(m)

Time (Predictor) ⇒ O(depth of tree)

Time (train) = (m log m and (d features) + m d)

= (m d log m)

= m × m × d × log m

Time - multi - class

→ $\frac{m}{k}$ (m/k)

↳ pick log(k)

Longest

cat 40

dog 30

Horse 25

Interpretability

DT → if - else → simplest